

Class 30 Geometry of chaotic attractors

Teams 5 and 6: Post photos of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

Questions

1. (The Rössler system)

Prof Strogatz describes the Rössler system as trying to capture the stretching and folding from a machine making taffy. The variables do not have specific physical meaning. The system is designed to provide a simple model of stretching and folding in phase space.

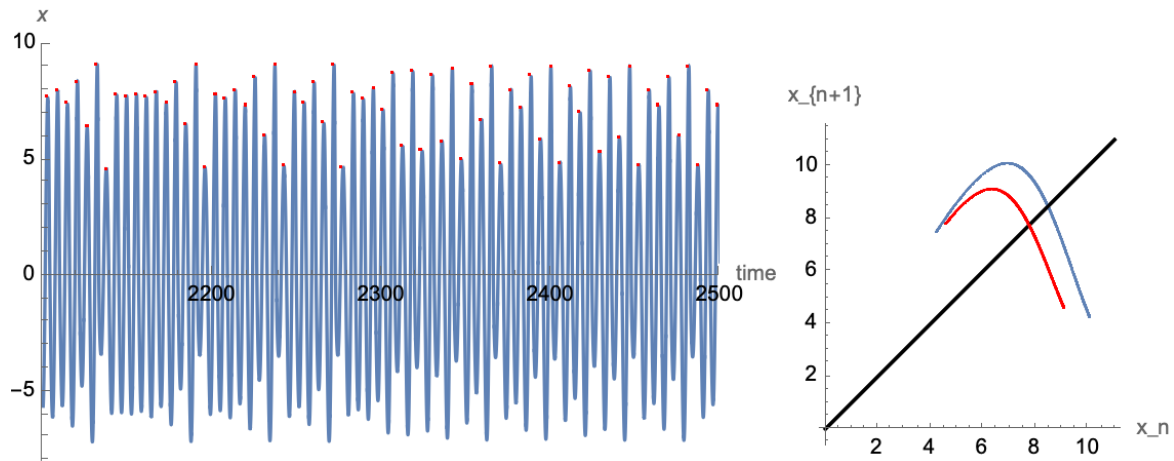
The Rössler system is

$$\begin{aligned}\dot{x} &= -y - z \\ \dot{y} &= x + ay \\ \dot{z} &= b + z(x - c).\end{aligned}$$

This system is slightly simpler than the Lorenz system, with slower volume contraction.

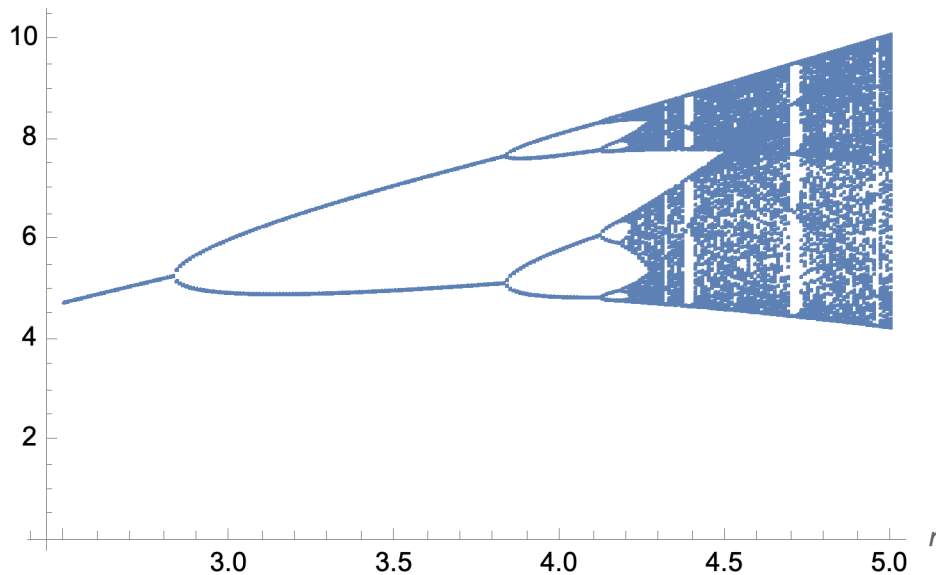
The sketches at the end of the handout, from Abraham and Shaw (1983), illustrate how trajectories in the Rössler system are stretched and folded in phase space.

- (a) Find the nonlinear term(s) in the system above.
 - (b) In the illustrations, imagine starting with a thin band of initial conditions. Where does the band appear to stretch under the flow?
 - (c) Where does that band appear to fold or wrap back?
 - (d) What does this stretching do to small differences in initial conditions?
 - (e) If this stretching and folding repeats many times, what kind of layered structure would you expect to develop? Compare this to the way $S_1, S_2, S_3, \dots$ approach the Cantor set.
2. We reduce the Rössler flow to a discrete sequence by recording successive local maxima of $x(t)$.
- (a) How is this related to a Poincaré map?
 - (b) What information does this keep? What does it discard?
 - (c) On the left is a trajectory with local maxima marked in red. On the right, each maximum is plotted against the next one (for $c = 4.5$ in red and $c = 5$ in blue).
These plots are not exact maps, since they have some width. Does a one-dimensional map seem like a reasonable model for what you see?

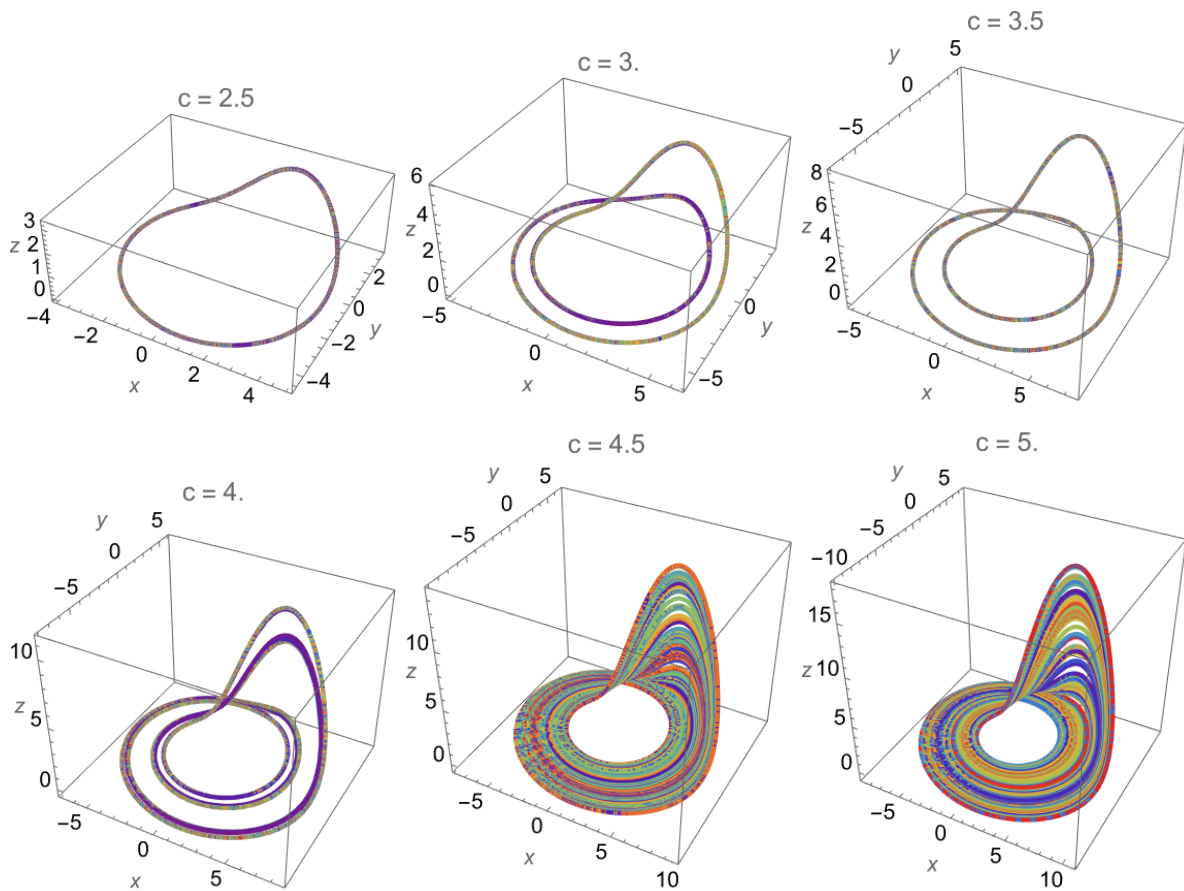


3. (varying a parameter) For each value of c , I compute a long trajectory and plot the local maxima of $x(t)$ against c (using later times in the trajectory).

orbit points



- What features of this orbit diagram look similar to the logistic map orbit diagram?
 - Where do you see periodic behavior?
 - Where do you see period-doubling?
 - Where do you see aperiodic or chaotic-looking behavior?
4. Here are some trajectories of the Rössler system at various values of c . You can hopefully see periodic structures from the orbit diagram appearing as trajectories in phase space.



- (a) What periods do you see? Which trajectories seem aperiodic?
 - (b) How does the geometry of the trajectory change as the period increases?
 - (c) How do these phase-space pictures reflect the structures you saw in the orbit diagram?
5. How are ideas from the maps phase of our course appearing in the Rössler system?
- You might think about return maps, stretching and folding, period-doubling, orbit diagrams, and fractal structure.*

